

Exam History of Science

Date: Wednesday May 31st, 2024

Lecturer: dr. D.J. Beckers

Time: 2h 15min

Course code: FEW X 400318

In general:

- (1) Make sure your name and student number are on ALL your work.
- (2) Answer in English sentences. Spelling and grammar should be correct either by English or by US standards. Just a few buzzwords never constitute an answer. Always explain yourself. Use appropriate examples to illustrate your answer.
- (3) This exam consists of 8 A-questions, 12 B-questions, 6 C-questions and a NO BONUS question. The A-questions are about the lectures; the B-questions refer to information from the book by Campbell-Kelly. The C-questions require you to combine the information from the book and lectures in your answer.
- (4) You pick four questions: 1 A-question, 2 B-questions and 1 C-question. Each question is worth 2 points. Indicate clearly which questions you're answering by mentioning either the number and / or the title of the question in your answer. If you answer more A-questions, only the first will be considered. Likewise for the B- and C-questions.
- (5) If you're *not* entitled to the bonus point, you additionally do the NO BONUS question. This question is worth one point.

A-Questions:

1. Universal language

During the lecture, the late 19th, early 20th century ideas about a universal language was connected to other attempts to understand and facilitate learning processes or human behaviour. Explain the line of thought, and describe how these ideas might be connected to post World War II ideas in cybernetics.

2. Creating computers

Discuss a history of one of the "European dwarves". Make a comparison to the story of a "US dwarf".

3. Metaphor

Elaborate on the rise of the 1950s metaphor of "electronic brain" to refer to computers. To what extent was the metaphor accurate. In your essay, mention specific computers to explain your line of thought.

4. AI in education

In what ways was AI instrumental in 1960s/70s education? Elaborate on two specific examples!

5. Science fiction

In the 1960s and 70s, (science fiction) novels and movies were instrumental in shaping the popular view on computing and AI – more so than the machines that existed in reality. Explain that remark. Discuss specific examples.

6. Internet

In the lecture the history of the rise of Internet was disconnected from the history of computing. Explain the line of thought and provide two examples to illustrate the explanatory value of the lecturer's view.

7. Cathy O'Neil

To what extent does Cathy O'Neil's work reflect on recent AI practice?

8. Stochastic parrots

Discuss the point of view of Timnit Gebru (et.al.) with respect to large language models.

B-Questions:

1. EDVAC

To what extent does Campbell-Kelly claim that the modern computer rose with the machines of “type EDVAC”?

2. Computer manufacturers

The reconstruction of the computer to a business machine in the early 1950s was performed by three types of firms. Describe the types of firms, and what it was that they actually reconstructed.

3. IBM

Campbell-Kelly discusses the rise of IBM in the computer industry by the example of the computer models they produced, mentioning the “sinister nickname” of Big Blue the company obtained by the end of the 1950s. Discuss the rise of IBM. In your story, explain both the “big” and the “blue” and why it was sinister.

4. Real time

Campbell-Kelly names his chapter on the rise of real time computing “Reaping the Whirlwind”. Discuss two possible meanings the authors may have with this chapter title.

5. Unbundling

Discuss IBM's unbundling decision (december 1968). Explain the reasons for the decision and some of the direct consequences.

6. Robert Noyce

Campbell-Kelly discusses the career of Robert Noyce as prime example for the shift of the centre of the US computer industry from East to West Coast. Explain the parallel. Discuss at least two reasons.

7. Radio

Which parallels does Campbell-Kelly suggest between the rise of domestic radio and the rise of personal computers?

8. Broadening the appeal

According to Campbell-Kelly, the personal computer industry in the 1980s was characterized by “meteoric growth and its almost total disconnection from the existing software-products industry”. Explain that remark. In your essay, mention the main firms relevant to the remark.

9. Vanevar Bush

Campbell-Kelly discusses the memex by Vanevar Bush, because it was instrumental in one of the dreams behind network technology. Explain how the 1945 memex was traditional, contemporary and futuristic at the same time.

10. Foxconn

Campbell-Kelly uses Foxconn as exemplar for one particular development in the globalizing computer industry. Explain his line of thought.

11. Cloud computing

To what extent was cloud computing new? Or was it, as it was popularly dubbed, “time sharing for the 21st century”?

12. Computing and governance

In his final chapter, Campbell-Kelly discusses various ways in which 21st century computing became entangled with governance. Explain two of these in detail.

C-Questions:

1. Market push and market pull

Discuss examples of AI developments that resulted from market push and market pull respectively.

2. Mechanizing proof

Compare 1970s developments in computer languages to attempts to create automatic proofs and proof checkers for mathematical proofs. Use the concept of agendas to make your comparison.

3. Software crisis

Discuss the software crisis making use of the concept of conflicting agendas.

4. Internet

The Internet boom of the early 21st century was discussed from different points of view in the book and the lectures. Elaborate on two of the main differences.

5. Defining technology

Discuss two 1950s examples of cybernetics as a defining technology (in Bolter's sense).

6. Sherry Turkle

To what extent did the work of Sherry Turkle contribute to our understanding of "digital culture".

NO BONUS

Elaborate on the role of Margeret Mead in the history of AI.